Akoizo Original Practice - Form 2

Division B | Rocks And Minerals | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Equal axes and right angles describe the conventional…

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

A. Monoclinic system

B. Orthorhombic system

C. Cubic system

D. Triclinic system

Answer: _____

2. (2 points) A two-stage cooling clue is…

Unit: Igneous processes, textures, and Bowen’s series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

A. Metallic luster

B. Perfect sorting

C. Porphyritic texture

D. One uniform streak

Answer: _____

3. (2 points) Best reference design includes…

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

A. Teacher names

B. Only specimen prices

C. Confusions and diagnostic distinctions

D. Only one photograph

Answer: _____

4. (2 points) Olivine contains…

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

A. No silicon

B. Isolated tetrahedra

C. Only metal atoms

D. A pure carbonate lattice

Answer: _____

5. (2 points) Cementation is…

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

A. Only breaking crystals

B. Binding grains by precipitated material

C. Moving grains downstream

D. Melting a whole rock

Answer: _____

6. (2 points) The best identification uses…

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

A. Several diagnostic observations

B. Color alone

C. One nickname

D. Only specimen size

Answer: _____

7. (2 points) Ore status depends partly on…

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

A. Economic extraction conditions

B. Color alone

C. Crystal size only

D. One fixed universal value

Answer: _____

8. (2 points) Nonfoliated means…

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

A. No strong planar fabric

B. Never metamorphosed

C. Always igneous

D. No pressure occurred

Answer: _____

9. (4 points) Why is slow cooling insufficient to predict size?

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

Answer: _____
_____
_____

10. (4 points) Why separate texture and composition?

Unit: Igneous processes, textures, and Bowen's series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

Answer: _____
_____
_____

11. (4 points) Design a calcite-quartz comparison.

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

Answer: _____
_____
_____

12. (4 points) Why does crystal elongation not prove chain structure?

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

Answer: _____
_____
_____

13. (4 points) Why is quartz sandstone a rock?

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

Answer: _____
_____
_____

14. (4 points) Explain a density calculation.

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

Answer: _____
_____
_____

15. (4 points) Why distinguish surface alteration?

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

Answer: _____
_____
_____

16. (4 points) Differentiate mineral and rock cleavage.

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

Answer: _____
_____
_____

17. (4 points) Distinguish symmetry and habit.

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

Answer: _____
_____
_____

18. (4 points) What supports an igneous interpretation?

Unit: Igneous processes, textures, and Bowen’s series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

Answer: _____
_____
_____